# Genuine Multipartite Entanglement Analysis — Molecular System

## MOLECULE

Benzene  (C6H6)
Basis set    : cc-pVDZ
Charge       : 0
Multiplicity : 1
Total e-     : 42
n-body order : 5

## GROUND STATE ENERGIES

E0 (Ha)  : -230.75355805
E1 (Ha)  : -230.57826157
Gap (Ha) : 0.17529648
HF  (Ha) : -230.71878874

## CONVEX-ROOF METHODS

Methods run    : gcg
Restarts       : 8

C_LB  : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB  : eigendecomposition bound

## ACTIVE SPACE  CAS(6,6)

Frozen core orbs : 18
Frozen e-        : 36
Active e-        : 6
Active orbs      : 6
Spin-orbitals    : 12
Hilbert dim      : 400

## SPIN PROPERTIES

<S2>         : 0.000000
S            : 0.0000
<Sz>         : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

## GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

## DOUBLE OCCUPANCY

<D> total  : 2.961506
<D> / site : 0.493584

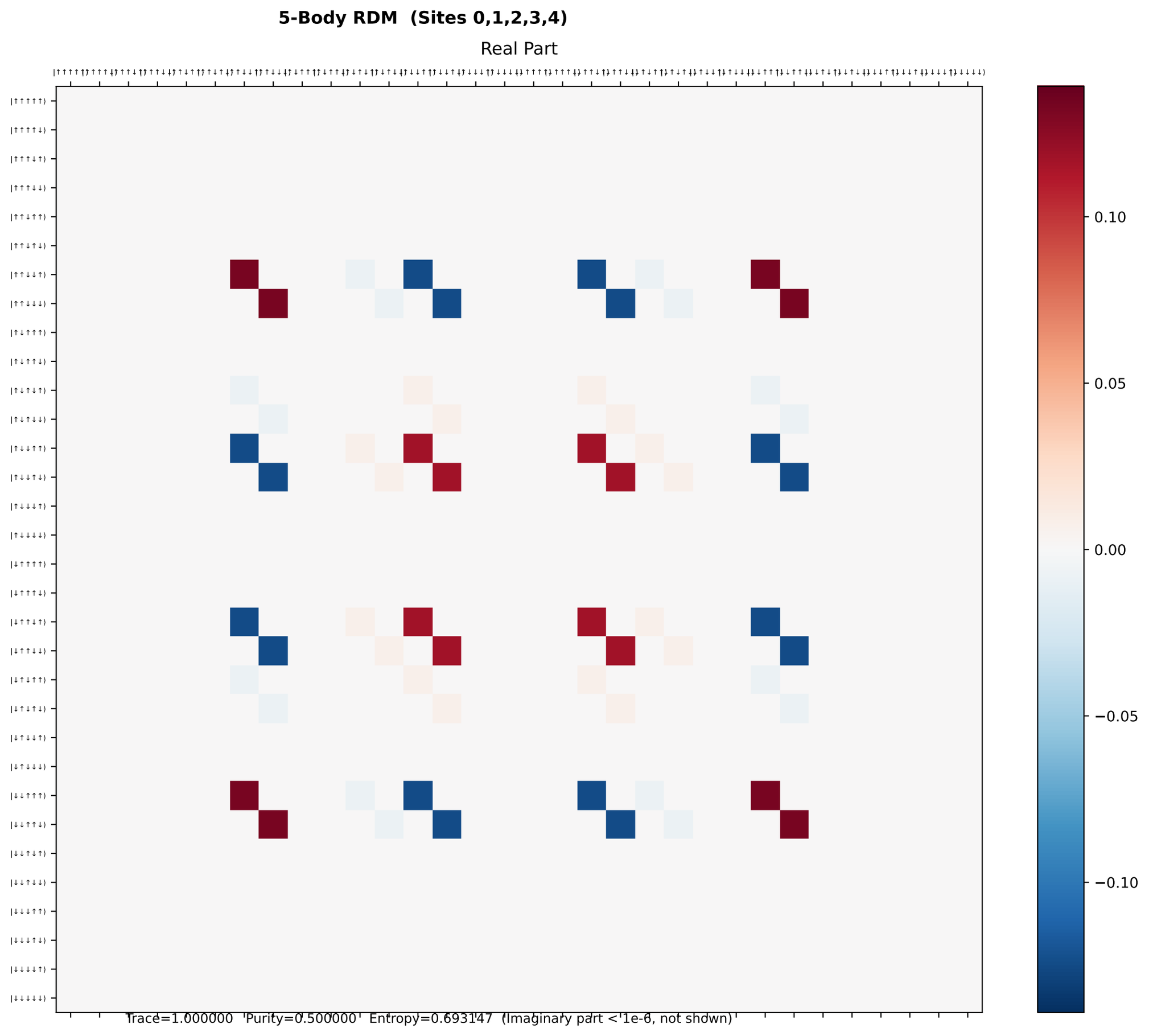
## THIS PDF

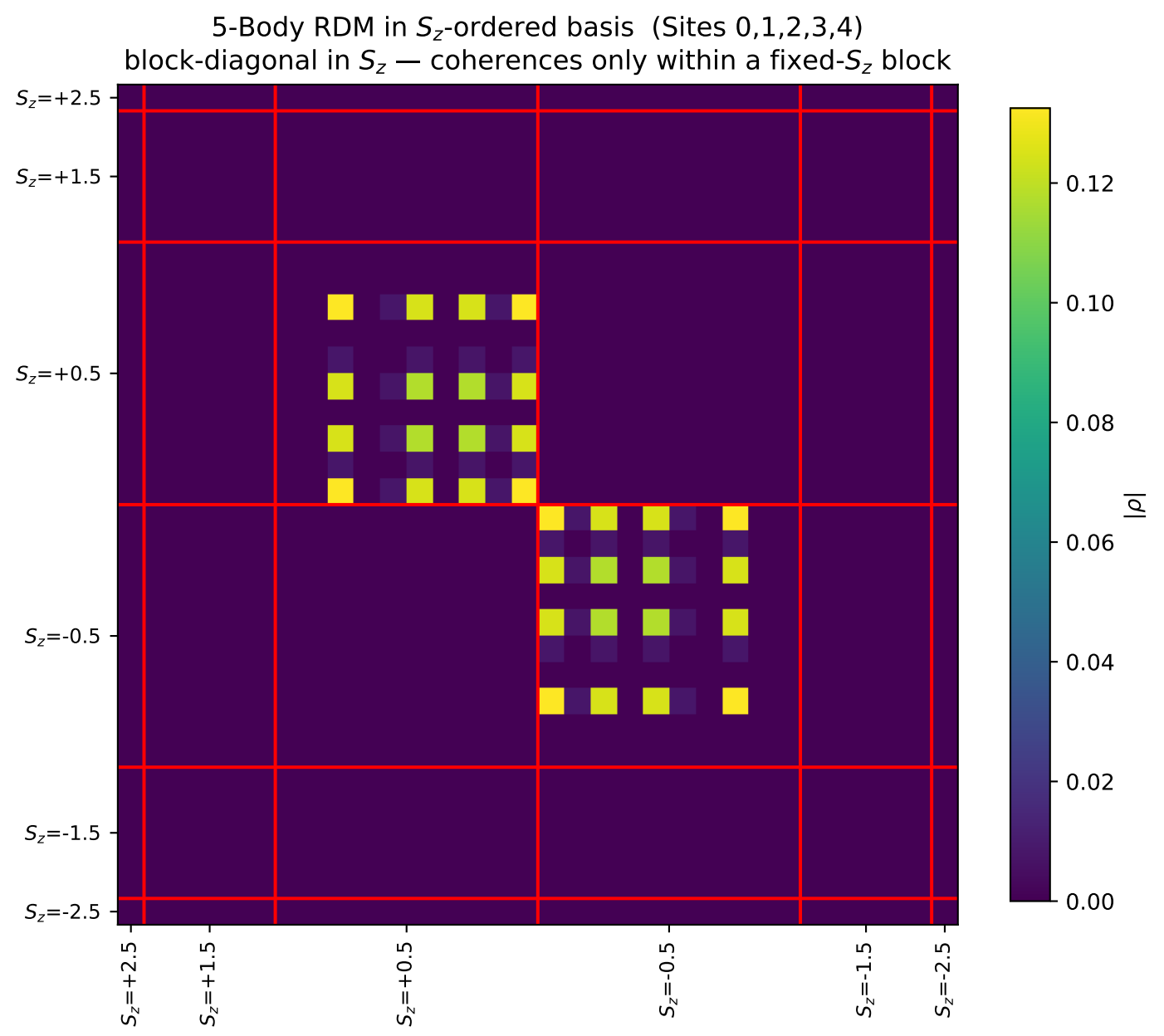
Body order : 5
Clusters   : 6

Date: 2026-07-10  23:30

## Summary — 5-Body RDMs  (Benzene, CAS(6,6))

| Orbitals | Purity | C_GME (pure) | C_LB | C_gcg | C_stiefel | C_UB | State |
|---|---|---|---|---|---|---|---|
| 0,1,2,3,4 | 0.500000 | — | — | — | — | — | MIXED |
| 0,1,2,3,5 | 0.500000 | — | — | — | — | — | MIXED |
| 0,1,2,4,5 | 0.500000 | — | — | — | — | — | MIXED |
| 0,1,3,4,5 | 0.500000 | — | — | — | — | — | MIXED |
| 0,2,3,4,5 | 0.500000 | — | — | — | — | — | WEAKLY MIXED |
| 1,2,3,4,5 | 0.500000 | — | — | — | — | — | MIXED |


5-Body RDM (Sites 0,1,2,3,4)
Real Part
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part <1e-6, not shown)




Max |coherence between different $S_z$| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate $S_z$ sectors.

# Analysis Summary - 5-Body RDM (Sites 0,1,2,3,4)

RDM Metrics
==========

| | |
|---|---|
| Trace(rho) | : 1.0000000000 |
| Purity Tr(rho^2) | : 0.5000000000 |
| Von Neumann entropy | : 0.6931471806 |
| Eigenvalue min | : -6.418492e-17 |
| Eigenvalue max | : 5.000000e-01 |
| Imaginary part | : < 1e-6 (not shown) |

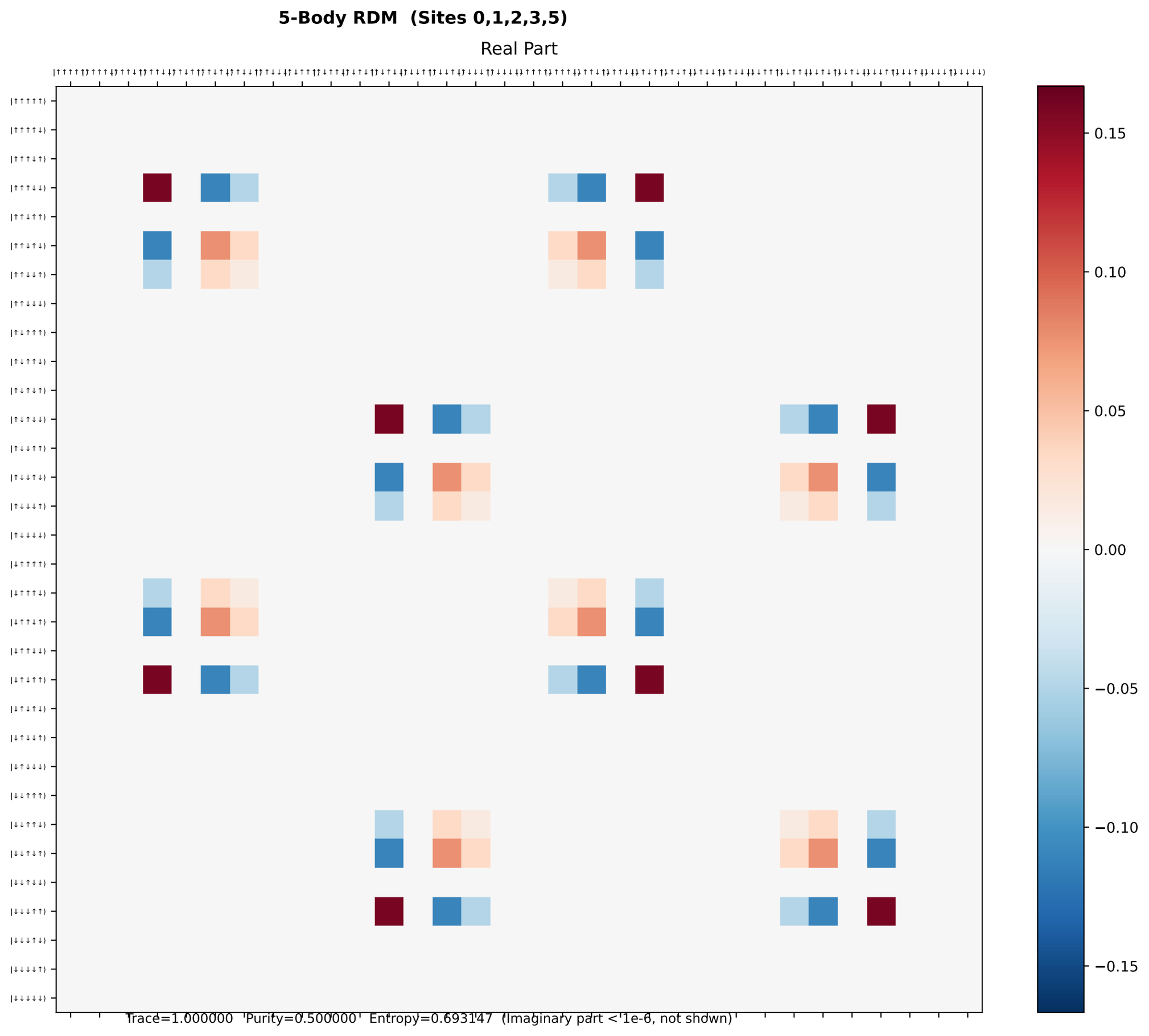
GME Analysis
==========

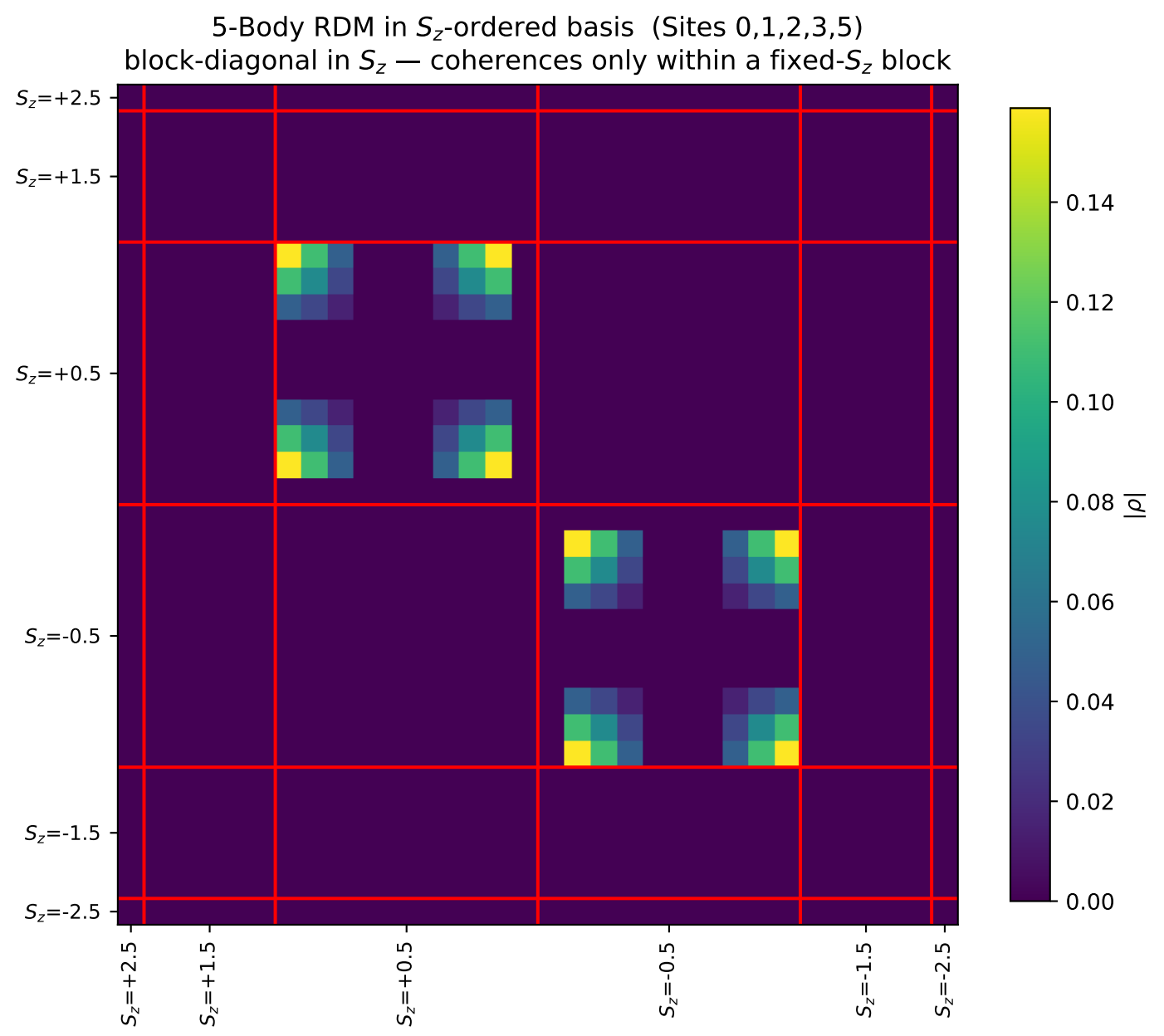
gamma = Tr(rho^2) = 0.50000000

| A | B | Tr(rho_A^2) |
|---|---|---|

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


5-Body RDM (Sites 0,1,2,3,5)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part <'1e-6, not shdwn)




Max |coherence between different $S_z$| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate $S_z$ sectors.

# Analysis Summary - 5-Body RDM (Sites 0,1,2,3,5)

RDM Metrics
==========

| | |
|---|---|
| Trace(rho) | 1.0000000000 |
| Purity Tr(rho^2) | 0.5000000000 |
| Von Neumann entropy | 0.6931471806 |
| Eigenvalue min | -8.057955e-17 |
| Eigenvalue max | 5.000000e-01 |
| Imaginary part | < 1e-6 (not shown) |

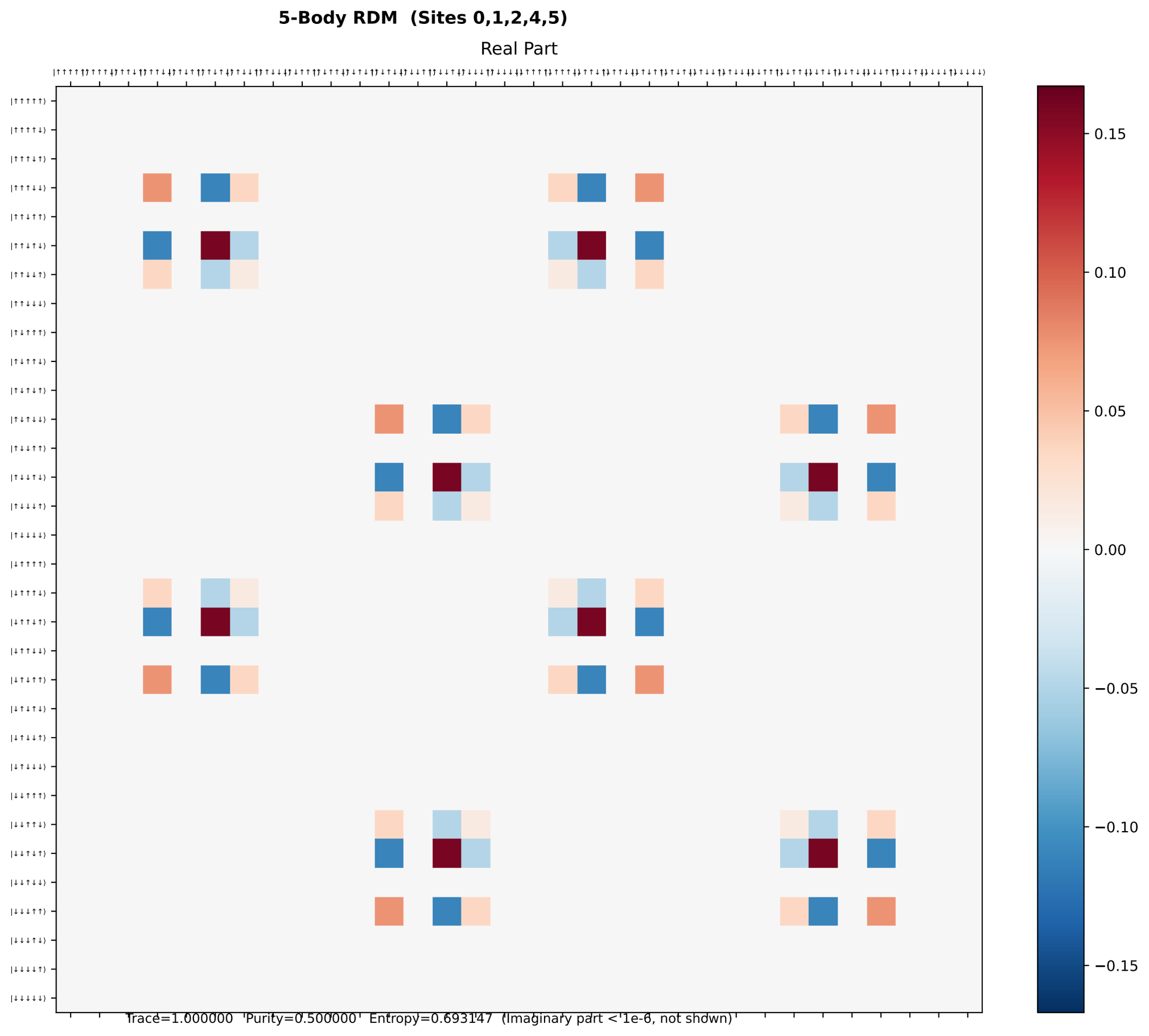
GME Analysis
==========

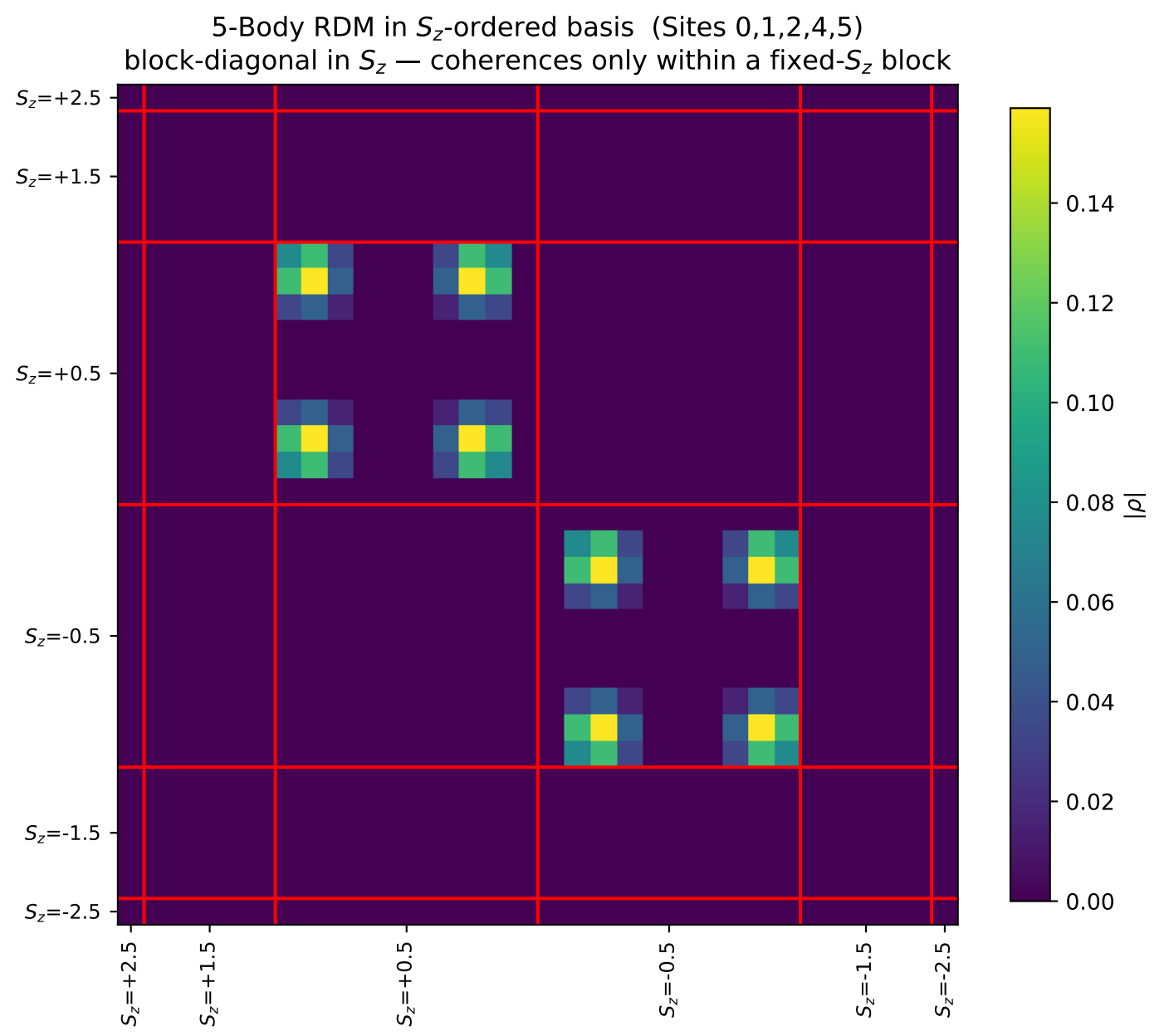
gamma = Tr(rho^2) = 0.50000000

| A | B | Tr(rho_A^2) |
|---|---|---|

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


5-Body RDM (Sites 0,1,2,4,5)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part <'1e-6, not shdwn)


5-Body RDM in $S_z$-ordered basis (Sites 0,1,2,4,5)
block-diagonal in $S_z$ — coherences only within a fixed-$S_z$ block
$S_z$=+2.5
$S_z$=+1.5
$S_z$=+0.5
$S_z$=-0.5
$S_z$=-1.5
$S_z$=-2.5
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$|\rho|$
Max |coherence between different $S_z$| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate $S_z$ sectors.

# Analysis Summary - 5-Body RDM (Sites 0,1,2,4,5)

RDM Metrics
==========

| | |
|---|---|
| Trace(rho) | : 1.0000000000 |
| Purity Tr(rho^2) | : 0.5000000000 |
| Von Neumann entropy | : 0.6931471806 |
| Eigenvalue min | : -2.517364e-17 |
| Eigenvalue max | : 5.000000e-01 |
| Imaginary part | : < 1e-6 (not shown) |

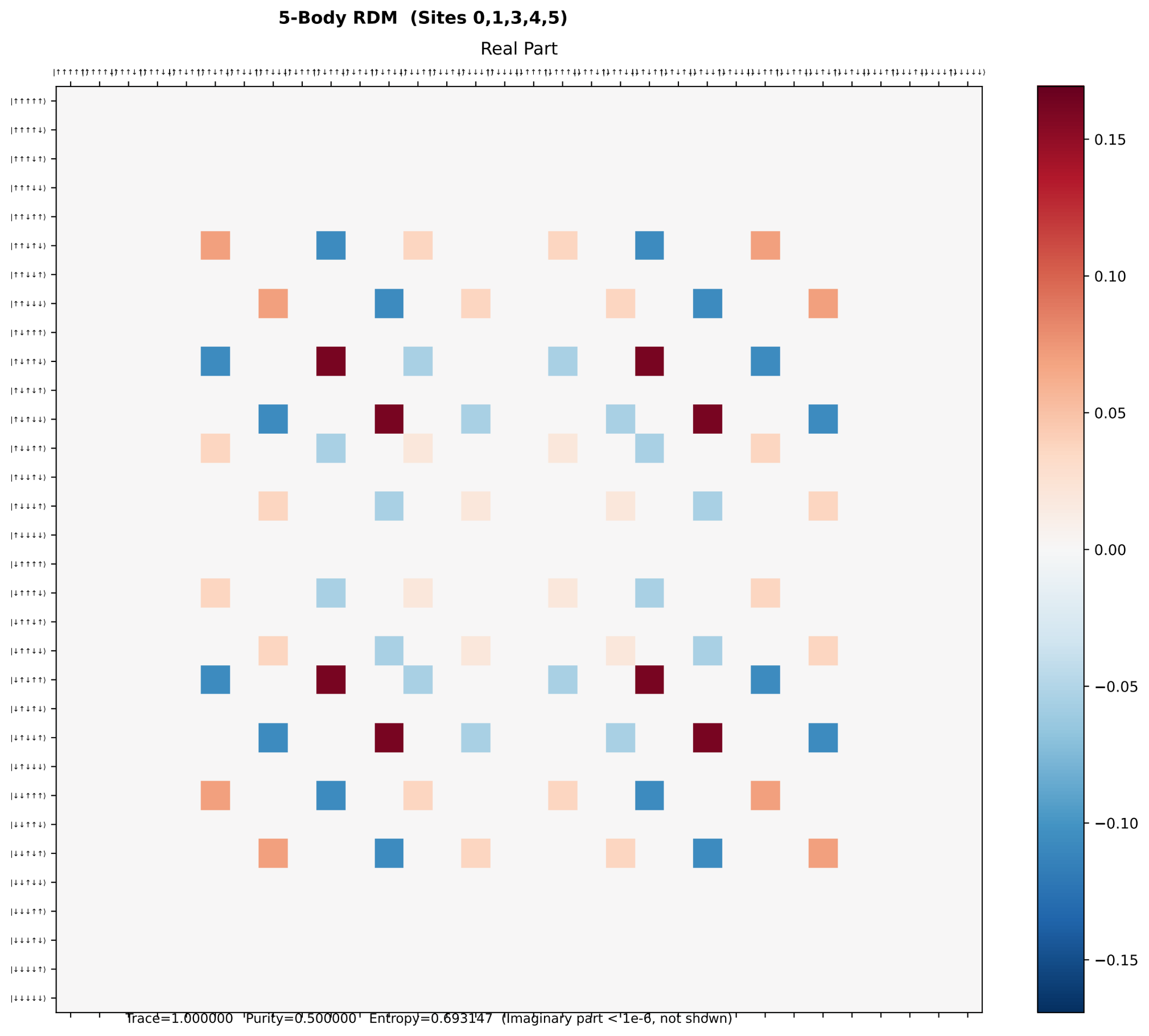
GME Analysis
==========

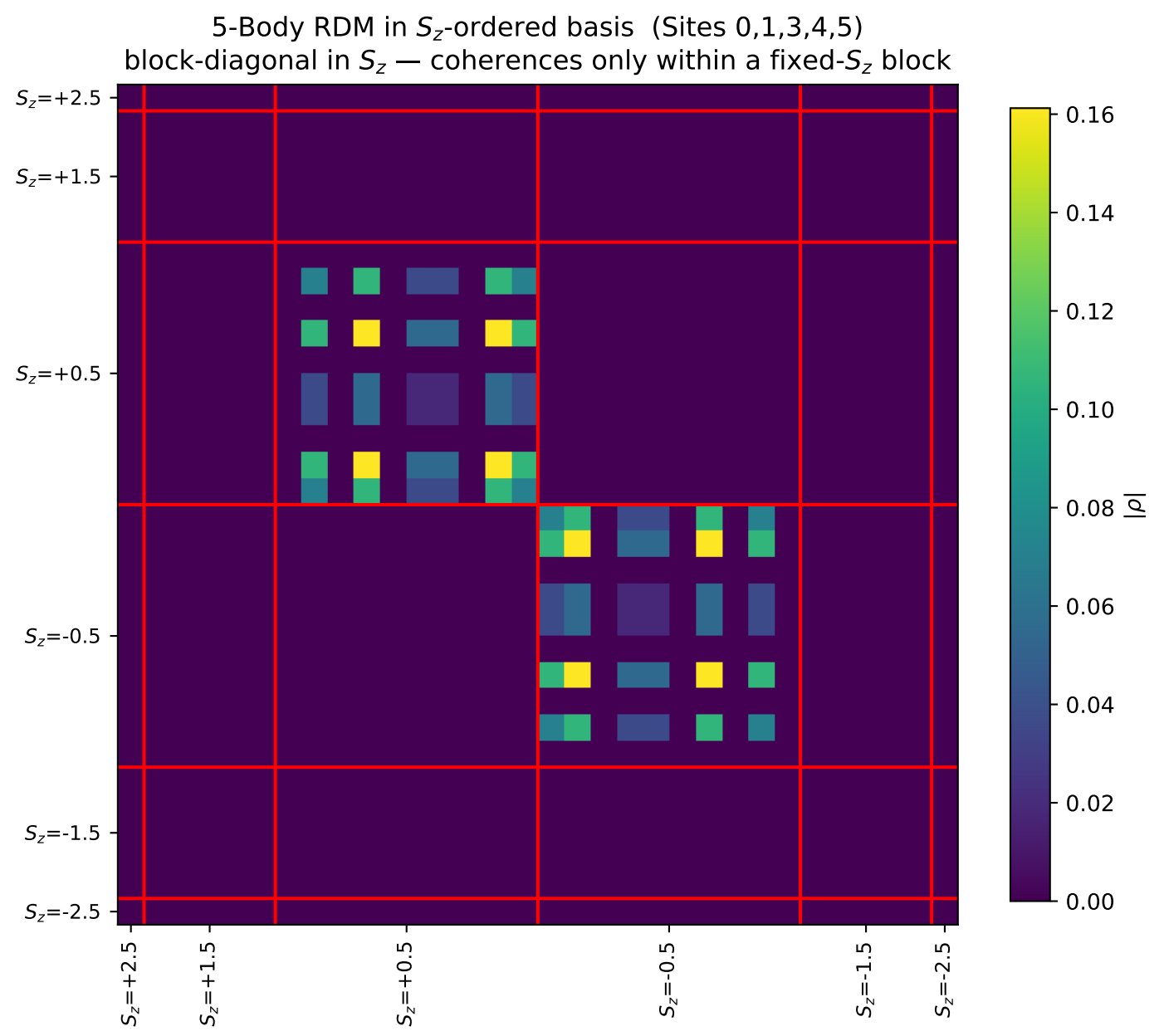
gamma = Tr(rho^2) = 0.50000000

| A | B | Tr(rho_A^2) |
|---|---|---|

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


5-Body RDM (Sites 0,1,3,4,5)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part <1e-6, not shown)




Max |coherence between different $S_z$| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate $S_z$ sectors.

# Analysis Summary - 5-Body RDM (Sites 0,1,3,4,5)

RDM Metrics
==========

| | |
|---|---|
| Trace(rho) | : 1.0000000000 |
| Purity Tr(rho^2) | : 0.5000000000 |
| Von Neumann entropy | : 0.6931471806 |
| Eigenvalue min | : -3.122941e-17 |
| Eigenvalue max | : 5.000000e-01 |
| Imaginary part | : < 1e-6 (not shown) |

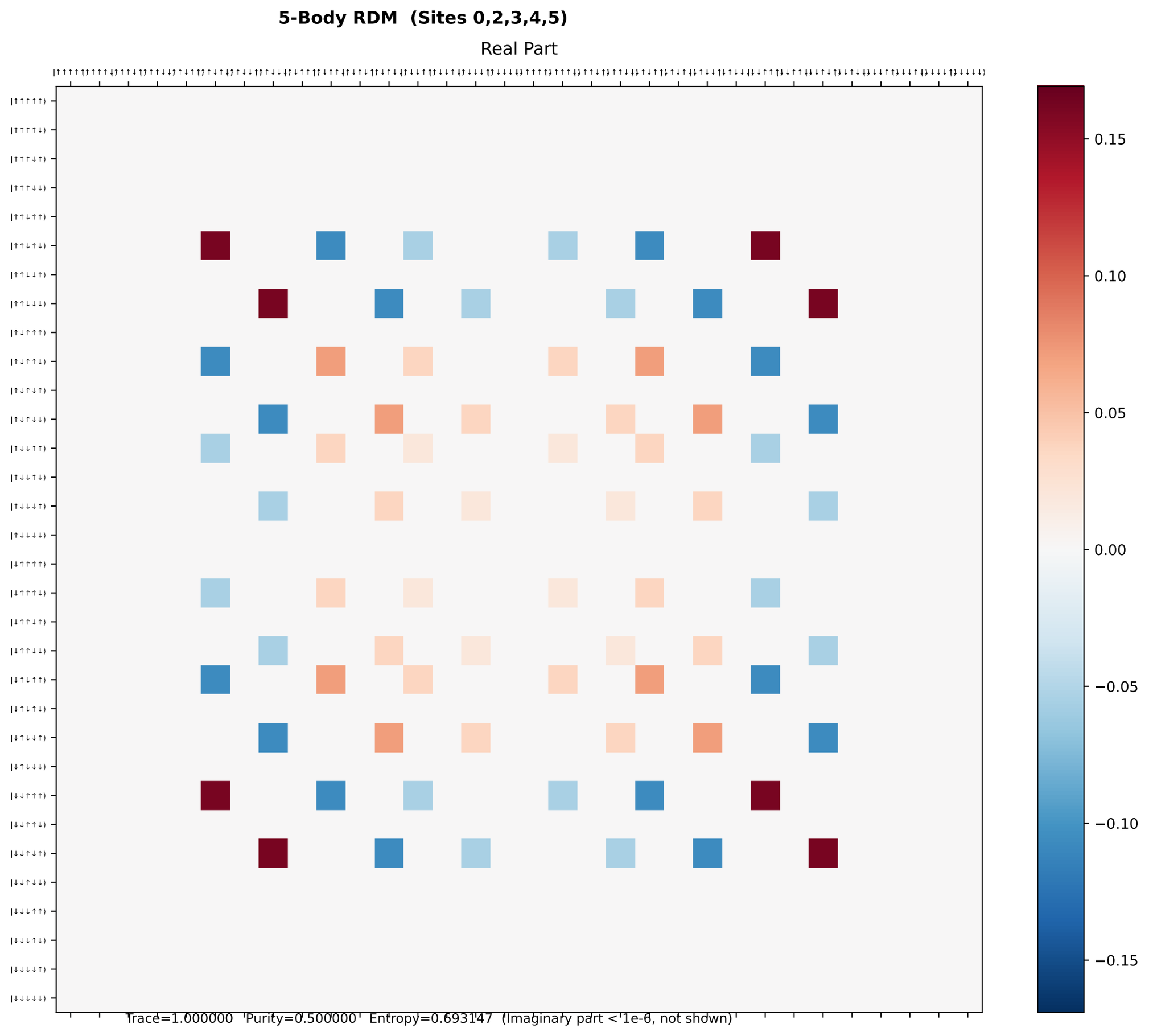
GME Analysis
==========

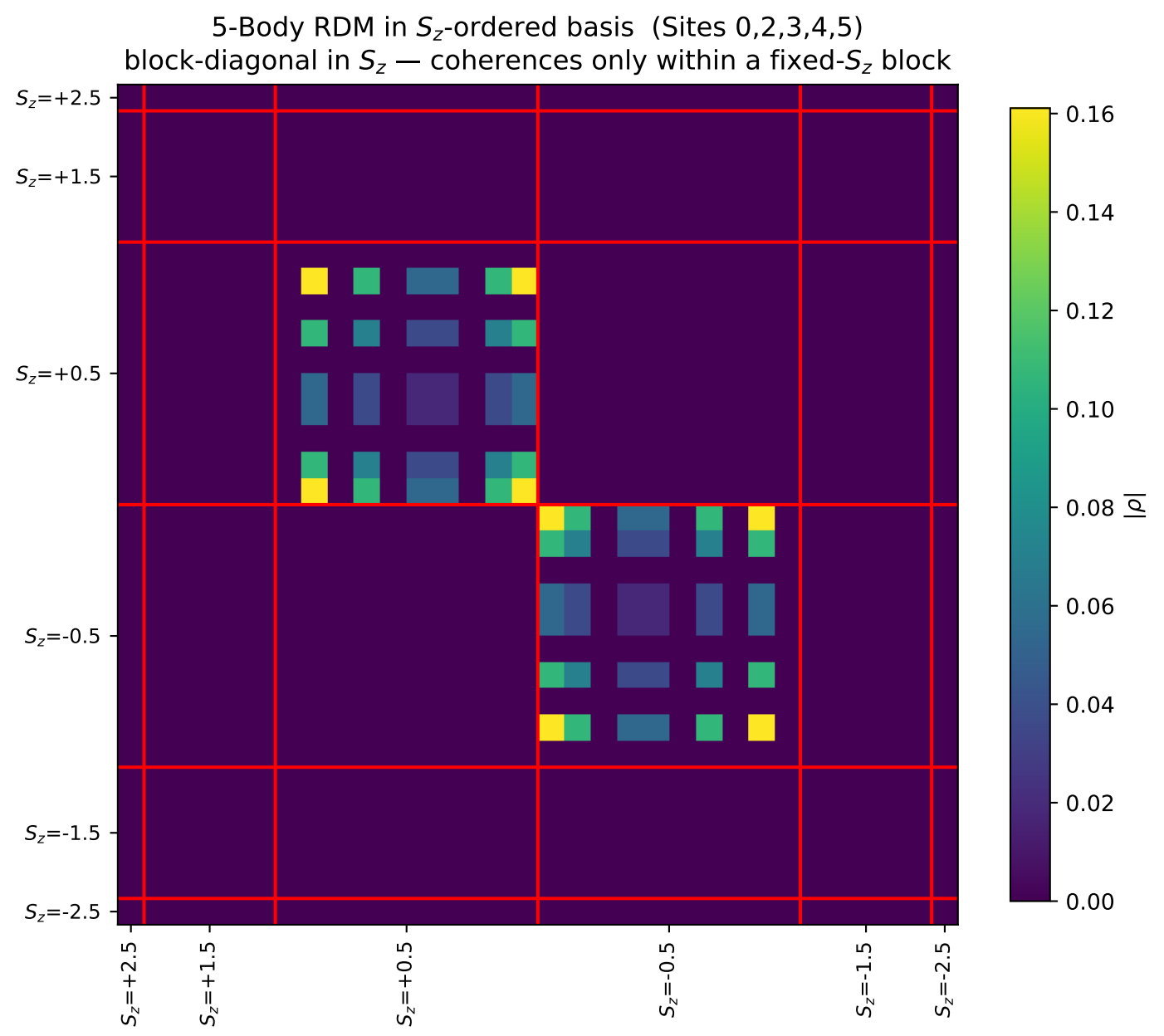
gamma = Tr(rho^2) = 0.50000000

| A | B | Tr(rho_A^2) |
|---|---|---|

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


5-Body RDM (Sites 0,2,3,4,5)
Real Part
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part <1e-6, not shown)




Max |coherence between different $S_z$| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate $S_z$ sectors.

# Analysis Summary - 5-Body RDM (Sites 0,2,3,4,5)

RDM Metrics
==========

| | |
|---|---|
| Trace(rho) | : 1.0000000000 |
| Purity Tr(rho^2) | : 0.5000000000 |
| Von Neumann entropy | : 0.6931471806 |
| Eigenvalue min | : -7.873478e-17 |
| Eigenvalue max | : 5.000000e-01 |
| Imaginary part | : < 1e-6 (not shown) |

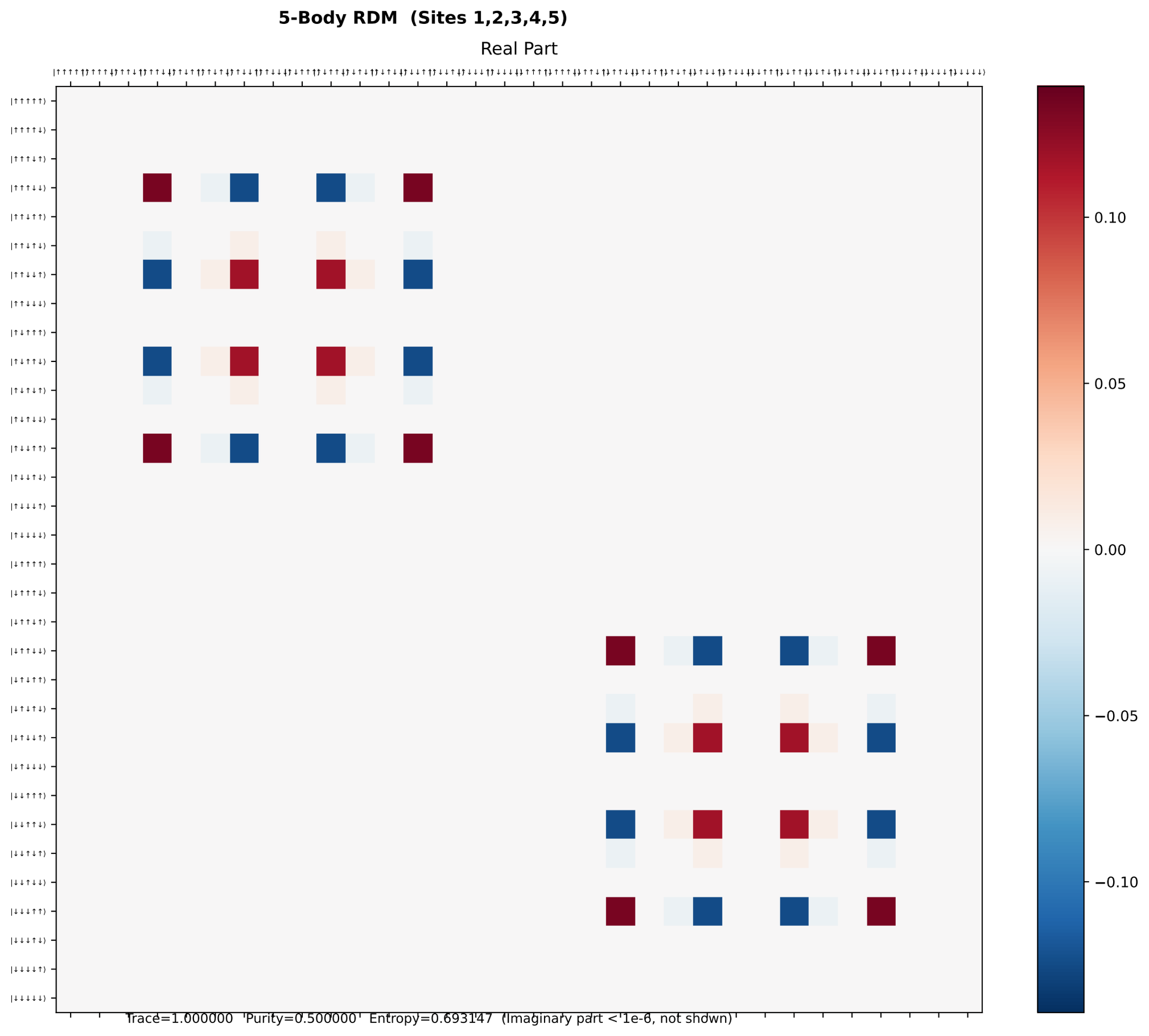
GME Analysis
==========

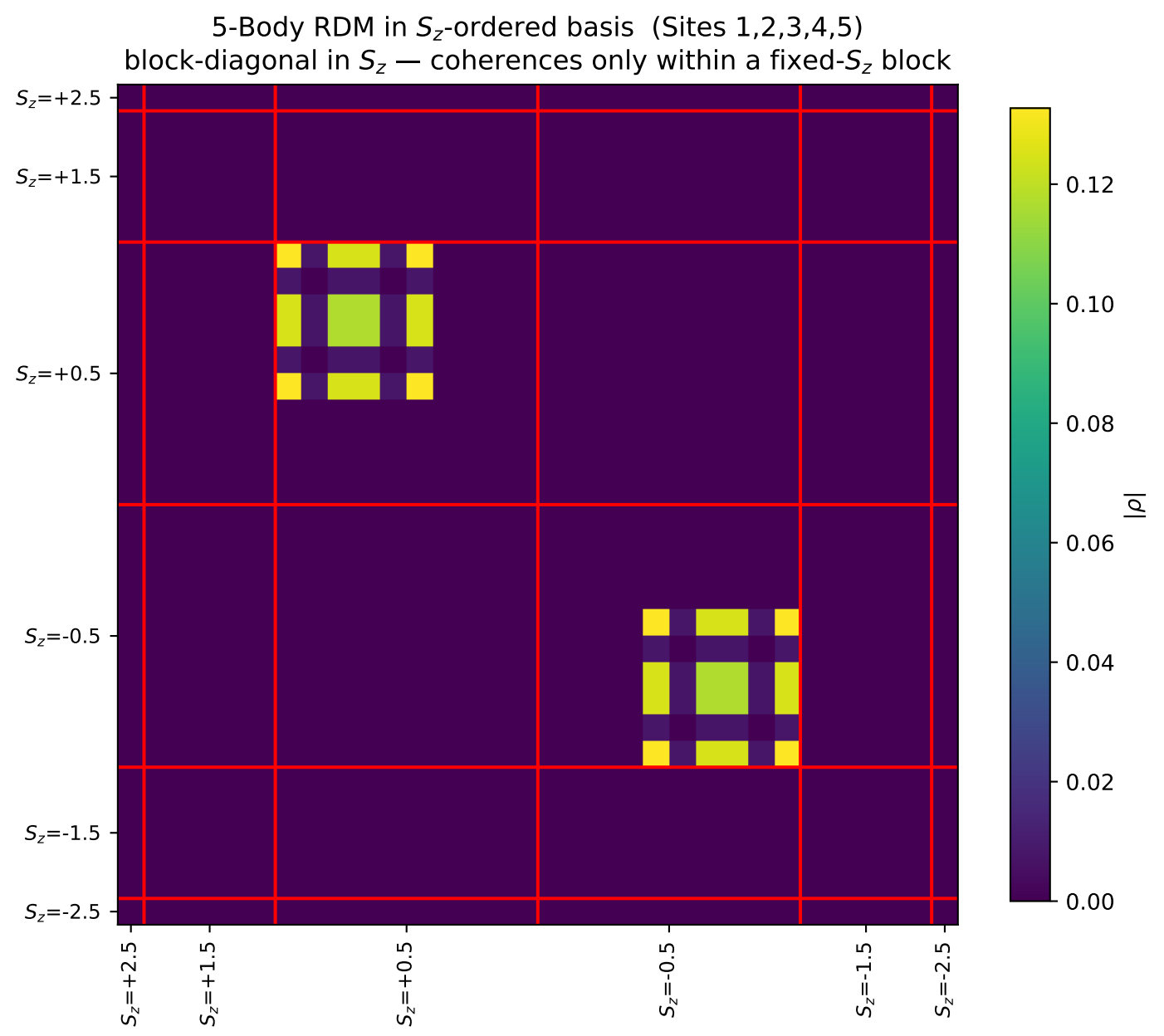
gamma = Tr(rho^2) = 0.50000000

| A | B | Tr(rho_A^2) |
|---|---|---|

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


5-Body RDM (Sites 1,2,3,4,5)
Real Part
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.500000 Entropy=0.693147 (Imaginary part <1e-6, not shown)




Max |coherence between different $S_z$| = 0.00e+00 ($\Rightarrow S_z$ conserved). Red lines separate $S_z$ sectors.

# Analysis Summary - 5-Body RDM (Sites 1,2,3,4,5)

RDM Metrics
==========

| | |
|---|---|
| Trace(rho) | 1.0000000000 |
| Purity Tr(rho^2) | 0.5000000000 |
| Von Neumann entropy | 0.6931471806 |
| Eigenvalue min | -1.339613e-16 |
| Eigenvalue max | 5.000000e-01 |
| Imaginary part | < 1e-6 (not shown) |

GME Analysis
==========

gamma = Tr(rho^2) = 0.50000000

| A | B | Tr(rho_A^2) |
|---|---|---|

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.